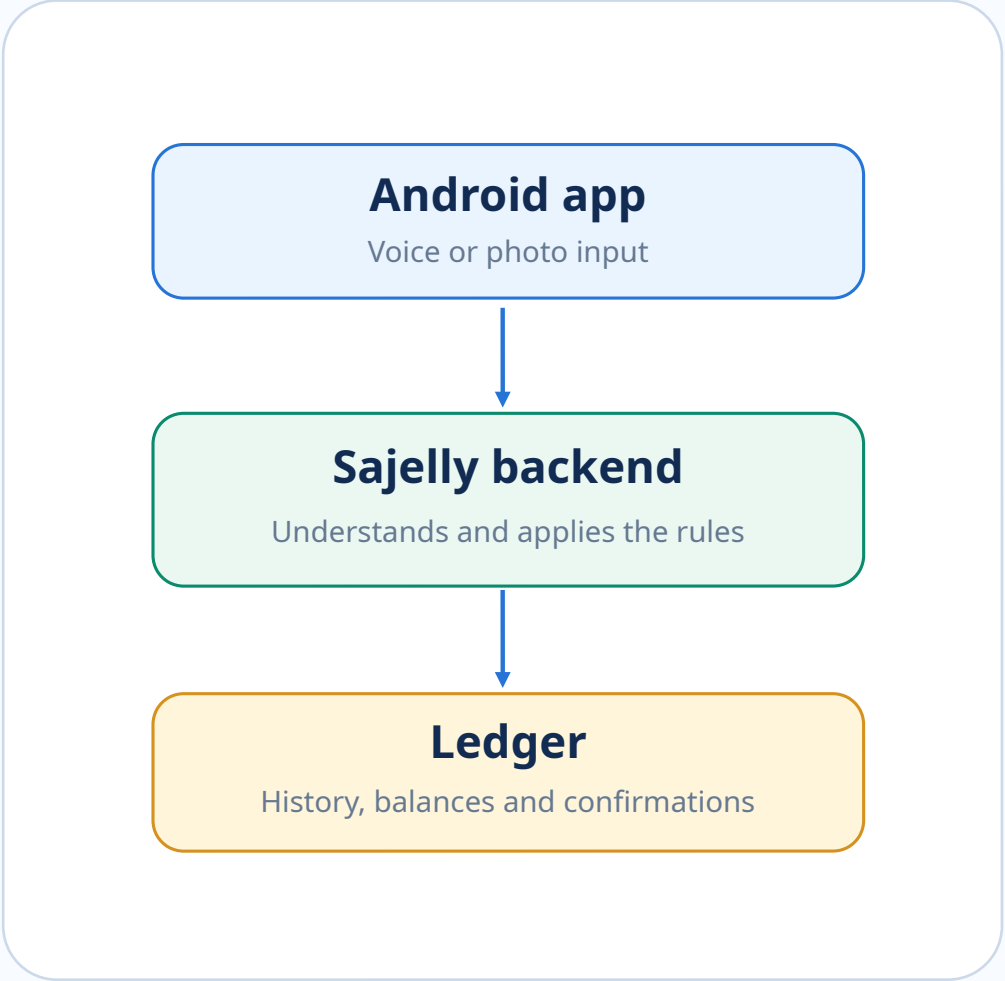
PRODUCT + SYSTEM ARCHITECTURE

سجلي
Sajelly

A voice accountant for a workshop

Listen. Understand. Ask. Confirm. Record.



From problem definition to a buildable system

01 NOV 2026

The problem is already understood

We are automating a job a family member used to perform.

Today: handwritten notebooks

The workshop sells ovens and other products. Customers pay in parts. The owner needs to know who owes money and how much.

The human helper

Listen to a question, find the right customer, calculate the balance, and ask when details are unclear.

Sajelly

Do the same job by voice in Egyptian Arabic. No reading, typing, or complex screens during normal use.

The product is not general accounting software. It is an assistant for customer debts and payments.

One screen. One conversation.

The Android app is a simple interface, not a place for financial decisions.



Speak

Press to record, press again to stop; upload one short clip.

Listen

The app receives audio back and plays it. Stop interrupts playback.

Photo

A small camera icon sends a notebook image through the same backend.

Login exists, but does not dominate daily use.

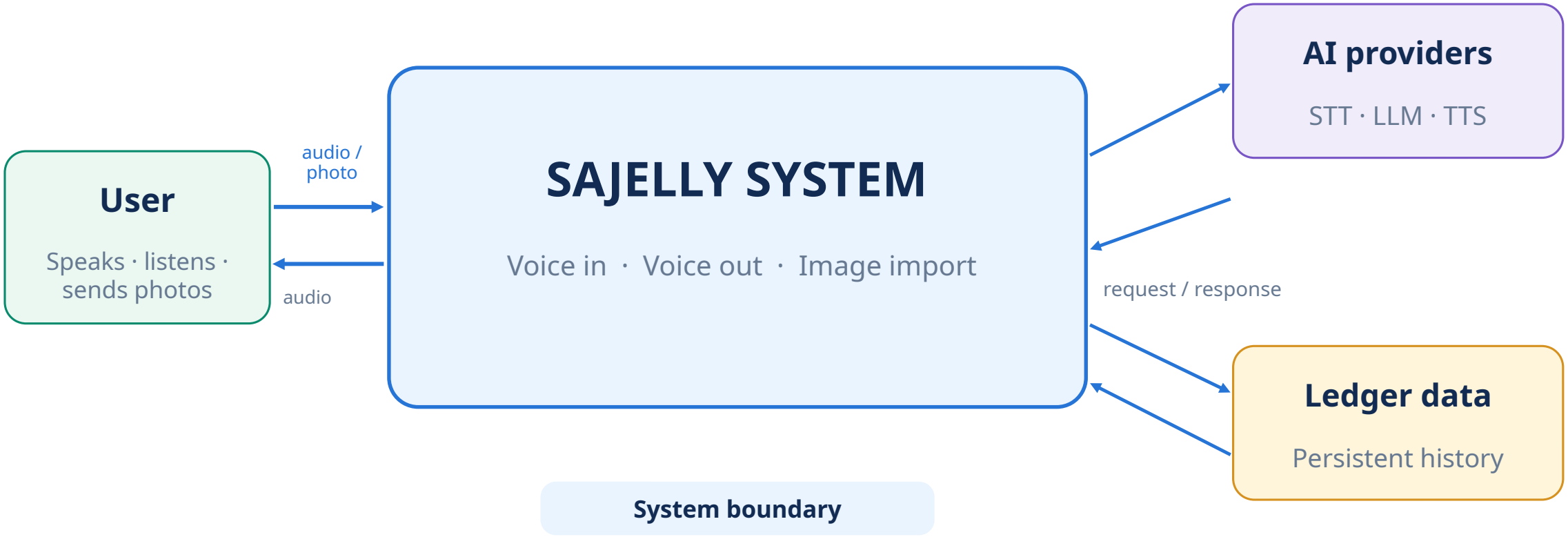
What Sajelly must do

Selected requirements, grouped by responsibility. IDs connect design to tests.

UX	**Speak, listen, photograph**	UX-002 · start recording UX-004 · stop spoken response
AG	**Understand and clarify**	AG-002 · ask when unclear AG-006 · do not invent details
FIN	**Answer using the ledger**	FIN-005 · customer balance FIN-006 · who still owes money
DATA	**Preserve financial history**	DATA-003 · balance from records DATA-004 · corrections keep history
SAFE	**Confirm before change**	SAFE-004 · link confirmation to proposal SAFE-006 · prevent duplicate commits

C1 · System context

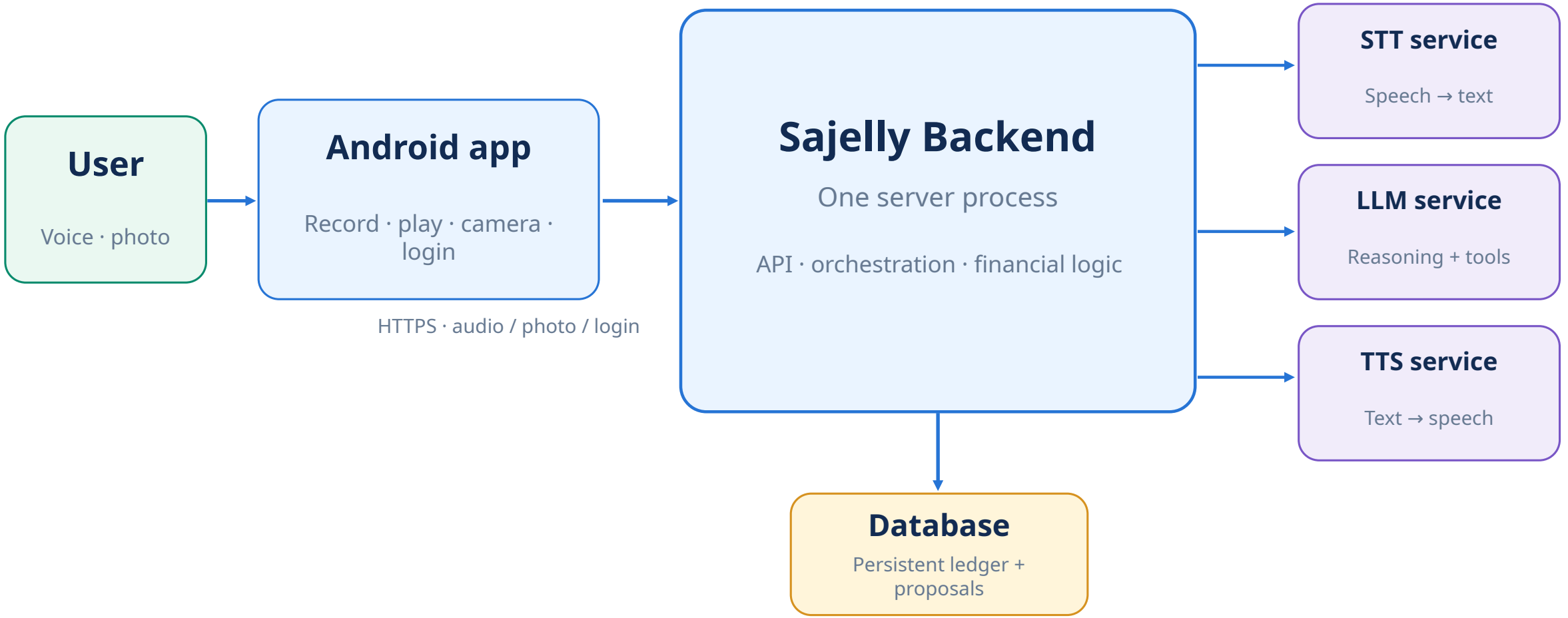
One user, one Sajelly system, external AI providers, and durable storage.



C1 intentionally hides Android, API endpoints, adapters and database internals.

C2 · Runtime containers

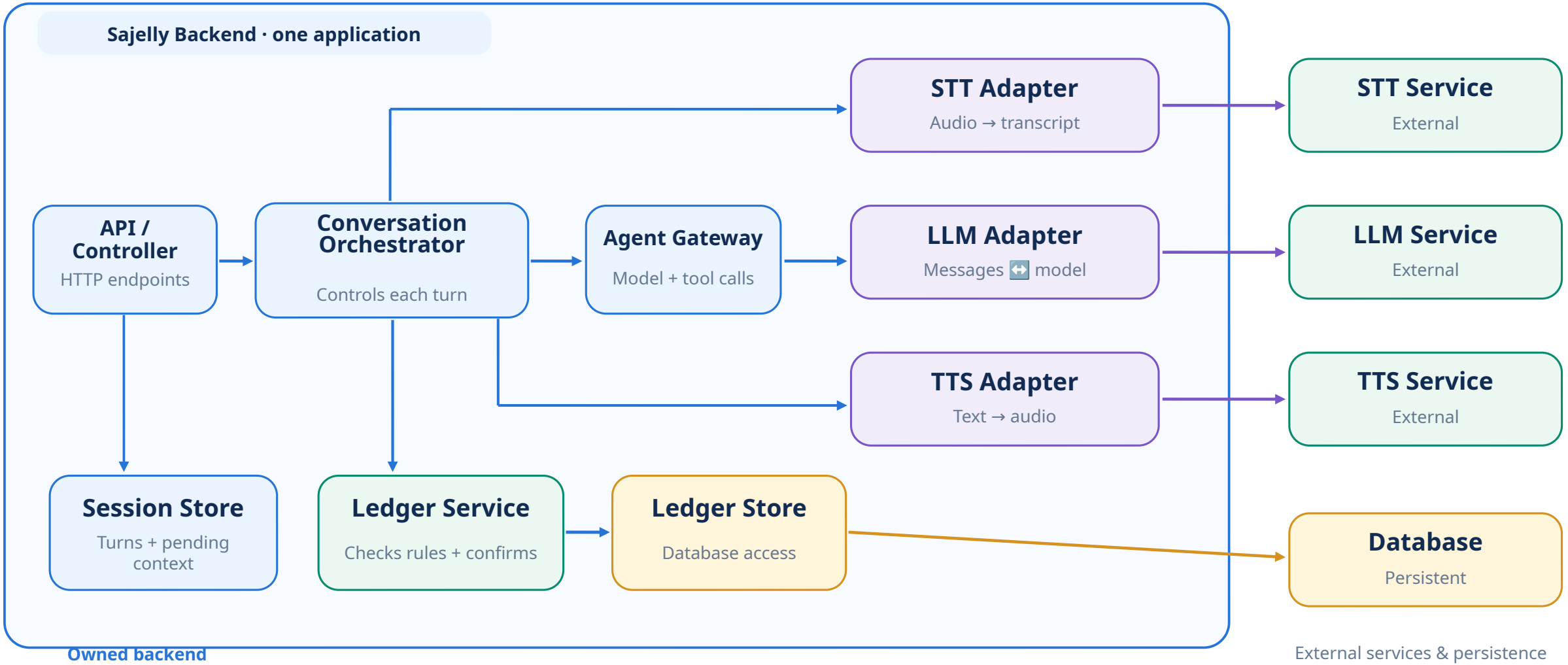
The pieces that run or store data, not separate roles for the same user.



Audio returned to the app; transcripts remain in the backend for inspection.

C3 · Sajelly backend components

The purple adapters connect one-to-one to their matching external services.



D1 · A spoken balance question

One complete user turn; no financial write takes place.

1 Android

Record, stop, upload audio

2 STT adapter

Send audio; receive transcript

3 Agent gateway

Ask LLM what action is needed

4 Ledger service

Query the customer balance

5 Agent gateway

Form a short Arabic answer

6 TTS adapter

Get spoken audio; return to app

Backend records transcripts, tool calls, results and diagnostic audio

Backend response to Android: AUDIO ONLY

D2 · A financial change requires confirmation

The assistant may interpret a reply, but it cannot directly change the ledger.

1 User speaks

“Ahmed paid 500 EGP.”

2 Agent clarifies

Find the intended customer.

3 Ledger proposes

Store fixed details + operation ID.

4 User confirms

A new spoken turn is linked to that ID.

5 Ledger commits

Validate identity, amount, confirmation, and duplicate protection.

6 Sajelly replies

Say “recorded” only after successful saving.

Safety rule: No matching, fresh confirmation → No ledger change. No commit success → No success message.

Build one vertical slice first

Do not implement the whole architecture at once.

First runnable slice

- Android: record / stop / play
- Backend: STT → agent → ledger query → TTS
- Known test customers and balances
- Short answer spoken in Egyptian Arabic

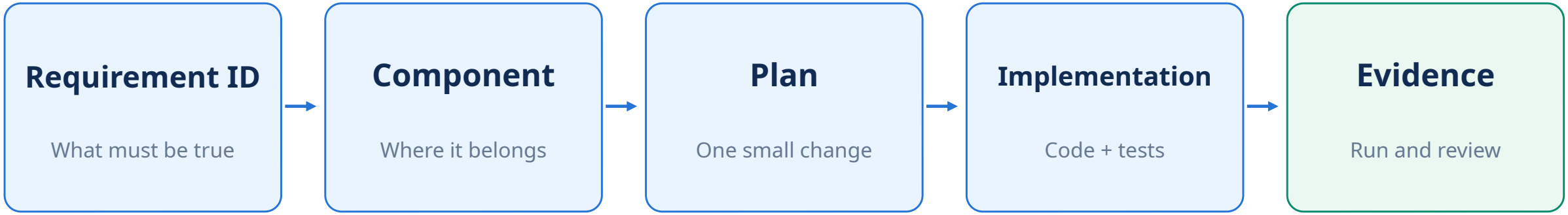
Definition of Done

- A complete voice question returns the right amount
- An ambiguous customer triggers clarification
- Stop interrupts spoken playback
- Audio and tool traces can be inspected

Next slices: confirmed writes · history corrections · notebook image import.

How agents build it without losing control

Requirements and architecture are inputs; passing tests is evidence.



Review decision

Accept when the Definition of Done passes; otherwise send the evidence back to the agent.

Architecture supplies boundaries. Tests verify behavior. The product owner accepts the result.